

Shorful Alam

Dhaka, Bangladesh | shorfulalamakib@gmail.com | +8801957189044 | [Portfolio](#) | [ORCID](#)

RESEARCH FOCUS

Bayesian inference and statistical methodology for sparse, structured data — empirical Bayes, hierarchical modeling, and small-area estimation. Current work uses R-INLA hierarchical models, comparing exchangeable and age-structured priors, to estimate adult mortality from DHS sibling survival histories, and I plan to extend this work toward spatial (CAR/ICAR) random-effects modeling and full MCMC-based estimation in Stan.

EDUCATION

M.Sc. in Statistics CGPA 3.46 / 4.0
Shahjalal University of Science and Technology, Sylhet, Bangladesh 2023 – 2024

B.Sc. in Statistics CGPA 3.40 / 4.0
Shahjalal University of Science and Technology, Sylhet, Bangladesh 2018 – 2023

Relevant Coursework: Probability, Theory of Statistics, Statistical Inference, Real Analysis, Stochastic Processes, Advanced Multivariate Analysis, Regression Analysis & Generalized Linear Models

PUBLICATIONS

Alam, S., Haque, N., Mazed, A., Ahmed, M. F. “Exchangeable versus Age-Structured Priors in Small-Area Estimation of Adult Mortality from Sibling Survival Histories.” **(First and Corresponding Author)**
Preprint: [SocArXiv](#) · Under review at Demography

Alam, S., Haque, N., Mazed, A., Mayen, M. S. A., Kowser, M. A., Hassan, M. “Network-Adjusted Empirical Bayes Estimation of Age-Specific Adult Mortality from DHS Sibling Survival Histories.” **(First and Corresponding Author)** Preprint: [Research Square](#) · Under review at Population Health Metrics

Alam, A.T.M.S., **Alam, S.,** Mobasshira, K., Hasan, M.N., Chowdhury, M.A.B., Uddin, M.J. “Exploring urban-rural inequalities of maternal healthcare utilization in Bangladesh.” Published · [Heliyon](#)

Mazed, A.*, **Alam, S.***, Ali, R., Uddin, M.J. “Health Literacy and Associated Factors among University Students in Bangladesh.” (*Co-first authorship) Forthcoming · BMC Public Health

Mazed, A.*, **Alam, S.***, Ali, R., Ahmed, N.U., Uddin, M.J. “Health Literacy Level among University Students: A Systematic Review.” (*Co-first authorship) Under review

Noor, S.T.A., Islam, R.B., Mostafa, M.S., **Alam, S.,** Islam, A.A. “Spatial Inequalities in Barriers to Healthcare Utilization Among Women in Bangladesh: A Nationwide Analysis.” Preprint: [Research Square](#) · Under review

Mayen, M.S.A., Mazed, A., Chowdhury, A., **Alam, S.,** Chowdhury, A.B., Uddin, J. “Examining the Prevalence and Determinants of Treatment-Seeking for ARI, Fever, and Diarrhea in Children Under Five in Bangladesh: A Pooled Analysis of DHS Data (2011–2022).” Under review

CONFERENCE & POSTER PRESENTATIONS

Treatment-Seeking Behaviors for ARI, Fever, and Diarrhea in Children Under Five in Bangladesh
8th International Conference on The Role of Statistics and Data Science in 4IR · 2025

Risk Factors of COVID-19 Pandemic's Footprints in Bangladesh: An Application of Negative Binomial Regression
International Seminar & Stat-Fest, SUST · 2023

RESEARCH EXPERIENCE

Independent Research — Adult Mortality Estimation, DHS Sibling Survival Data

Self-initiated; executed with BEpPh Research Team collaborators

- Conceived and led two Bayesian/empirical-Bayes research projects on adult mortality estimation, directing a small collaborator team through model development, validation, and manuscript preparation.
- Built R-INLA hierarchical models and heteroscedastic-SURE empirical Bayes shrinkage; validated model performance via Monte Carlo simulation under known ground truth.

Research Assistant — Health Literacy and Associated Factors among University Students in Bangladesh

SUST Research Center · Supervisor: Dr. Md Jamal Uddin · Jul 2022 – Jun 2023

- Co-developed the research proposal and conducted a PRISMA-guided systematic review, screening 9,390 records and identifying 10 eligible studies through dual-reviewer screening.
- Designed and administered a cross-sectional survey and analyzed results via multiple linear regression.
- Supervised junior students and contributed to two manuscripts and multiple conference presentations.

Research Group Member — BEpPh Research Team (Voluntary)

Faculty-led research group, Bangladesh · Jan 2022 – Present

- Collaborate on applied health research projects spanning biostatistics, epidemiology, and demography.

EMPLOYMENT

Assistant Teacher (Government Employee)

Ministry of Primary and Mass Education, Government of Bangladesh · June 2024 – Present

- Delivers curriculum-based lessons, monitors student progress, and maintains class records as class teacher.

TECHNICAL SKILLS

Bayesian & Statistical Methods: Hierarchical Bayesian modeling (R-INLA — IID, RW2, penalized-complexity priors) · Empirical Bayes (SURE-based heteroscedastic shrinkage) · Small-area estimation (Fay–Herriot models) · Survival analysis · Monte Carlo simulation · Design-based bootstrap variance estimation · Regression · Time-series analysis.

Statistical Machine Learning: Supervised and unsupervised learning (Random Forest · XGBoost · SVM · k-NN · Clustering · PCA).

Programming & Software: R (INLA, tidyverse, caret, surveybootstrap, ggplot2) · Python (NumPy, pandas, scikit-learn, Keras, Matplotlib, Seaborn) · SQL · Stata · SPSS · LaTeX.

Applied Research Areas: Demography · Public health · Health survey data · Spatial and inequality analysis.

STANDARDIZED TEST SCORES: IELTS (Nov 2025) — Overall 7.0 (L 6.5 · R 7.5 · W 6.5 · S 7.0)

REFERENCES

Dr. Md Jamal Uddin

Professor, Department of Statistics, SUST, Sylhet · jamal-sta@sust.edu · Website: [Dr. Md Jamal Uddin](#)

Dr. S. M. Khurshid Alam

Professor, Department of Statistics, SUST, Sylhet · khurshid@sust.edu · Website: [Dr. S. M. Khurshid Alam](#)